

Hermes Agent + Discord + OpenAI Realtime Voice Setup

PDF version of the setup guide. Full Markdown source:

website/docs/guides/hermes-discord-openai-realtime-voice-setup.md

Goal: run Hermes as a Discord bot that can join a voice channel and use OpenAI Realtime speech-to-speech.

Architecture:

Discord RTP/Opus -> PCM -> OpenAI Realtime WebSocket -> Discord voice playback.

For long work, Realtime calls a safe Hermes subagent handoff tool and posts durable output to the text channel.

Prerequisites:

1. Discord application and bot token from <https://discord.com/developers/applications>.
2. OpenAI Platform API key from <https://platform.openai.com/api-keys> with Realtime access.
3. Hermes Agent installed from package or source checkout.
4. ffmpeg, Opus, Python voice dependencies, and a running Hermes gateway.
5. Bot permissions: View Channels, Send Messages, Read History, Connect, Speak, Use Voice Activity.

Install basics:

macOS: brew install ffmpeg opus portaudio espeak-ng

Ubuntu/Debian: `sudo apt install ffmpeg libopus0 portaudio19-dev espeak-ng`

```
Python: python -m pip install -U "hermes-agent[messaging,voice]" "discord.py[voice]" "PyNaCl>=1.5.0" davey websockets
```

Configuration

Add secrets to ~/.hermes/.env. Never commit this file.

DISCORD_BOT_TOKEN=your-discord-bot-token

DISCORD_ALLOWED_USERS=your-discord-user-id

OPENAI_REALTIME_API_KEY=sk-...

OPENAI_REALTIME_MODEL=gpt-realtime

OPENAI_REALTIME_VOICE=alloy

OPENAI_REALTIME_INSTRUCTIONS=Respond in English only. Keep spoken replies concise. Use Hermes subagent ha

Hermes checks Realtime keys in this order:

1. OPENAI_REALTIME_API_KEY
2. VOICE_TOOLS_OPENAI_KEY
3. OPENAI_API_KEY

Recommended ~/.hermes/config.yaml fallback settings:

stt.provider: local

tts.provider: edge

voice.auto_tts: false

discord.require_mention: true

discord.auto_thread: true

Discord Developer Portal:

Enable Presence Intent, Server Members Intent, and Message Content Intent.

Invite with scopes bot and applications.commands. Verify voice permissions in the actual target channel.

Verification and Use

Verify layer-by-layer:

1. Start gateway: hermes gateway
2. Test text in Discord: /whoami or mention/DM the bot.
3. Run from checkout: python scripts/discord-voice-doctor.py
4. Run Realtime smokes if available:
python ~/.hermes/skills/devops/hermes-operations/scripts/openai_realtime_text_smoke.py
python ~/.hermes/skills/devops/hermes-operations/scripts/openai_realtime_audio_smoke.py
5. Test Hermes-native voice first: /voice join, /voice status, /voice leave.
6. Test Realtime voice: join a VC, then run /voice realtime in the linked text channel.

Useful Discord commands:

/voice realtime - join VC with OpenAI Realtime engine
/voice join - join VC with Hermes-native STT/TTS engine
/voice status - inspect state
/voice off - switch live voice engine back/off
/voice leave - disconnect and clear state

PASS means text, slash commands, Discord voice transport, OpenAI Realtime smoke, /voice realtime audio, and subagent.
PARTIAL means only some layers are verified. BLOCKED means a required key, token, permission, dependency, network